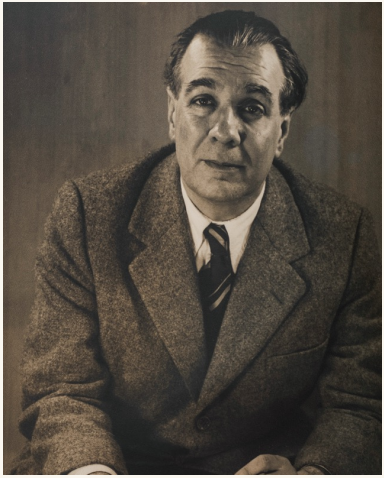
Two Engines

Memory, prediction, and the difference between them

Carlos Cotrini

ETH Zürich | AI in Industry

Week 2 — Spring 2026



"To think is to forget differences, generalize, abstract."

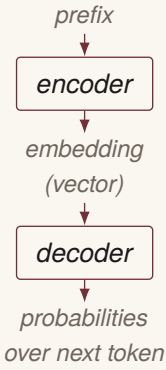
Jorge Luis Borges, *Funes the Memorious*, 1942

Last week, in language

You met the transformer. The summary we landed on:

An LLM ≈ encoder (builds meaning) + decoder (continues the sentence).

The *encoder* reads a prefix and produces an *embedding* — a vector summarising what the prefix says. The *decoder* takes that vector and writes a probability distribution over the next token.



Two halves of one machine

The encoder and decoder have older names.

- Encoder ≈ *compression*. A long sequence becomes a short vector.
- Decoder ≈ *prediction*. The vector becomes a bet about what comes next.

For ninety minutes today we will do each half explicitly — but for data that isn't language. *Customers, not words. Campaigns, not tokens.*

This framework is older than the transformer. It is from a 1948 paper by Shannon.

The promise

Machine learning is the engineering of two old theorems. *Compression* — the shortest faithful description of the past — and *prediction* — the best bet about the future — are two views of the same subject, governed by Shannon's 1948 theorems. We will build them in two halves and collapse them in the third.

"... a Map of the Empire whose size was that of the Empire, and which coincided point for point with it."

Jorge Luis Borges, *On Exactitude in Science*, 1946

Act I — Compression

The data: 21 readers, 16 books

A small public library has kept a quiet record.

For each of *21 anonymous readers* and each of *16 books*, one cell records whether the reader liked the book, disliked it, or has not yet read it.

Three symbols. Three hundred and thirty-six cells.

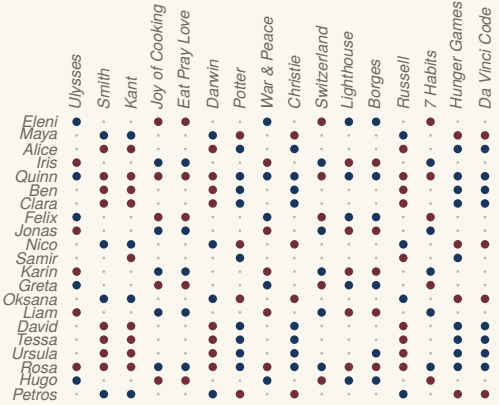
No names, no genres, no author photographs — only the matrix.

Sixteen books

Throughout the lecture each book is referred to by its *short label* (right column).

The Wealth of Nations	Adam Smith, 1776	*Smith*
Critique of Pure Reason	Immanuel Kant, 1781	*Kant*
On the Origin of Species	Charles Darwin, 1859	*Darwin*
War and Peace	Leo Tolstoy, 1869	*War & Peace*
Ulysses	James Joyce, 1922	*Ulysses*
To the Lighthouse	Virginia Woolf, 1927	*Lighthouse*
The Joy of Cooking	Irma Rombauer, 1931	*Joy of Cooking*
Murder on the Orient Express	Agatha Christie, 1934	*Christie*
Ficciones	Jorge Luis Borges, 1944	*Borges*
A History of Western Philosophy	Bertrand Russell, 1945	*Russell*
The 7 Habits of Highly Effective People	Stephen Covey, 1989	*7 Habits*
Harry Potter and the Philosopher's Stone	J. K. Rowling, 1997	*Potter*
The Da Vinci Code	Dan Brown, 2003	*Da Vinci Code*
Eat, Pray, Love	Elizabeth Gilbert, 2006	*Eat Pray Love*
The Hunger Games	Suzanne Collins, 2008	*Hunger Games*
Lonely Planet: Switzerland	travel guide, 2024	*Switzerland*

Raw data

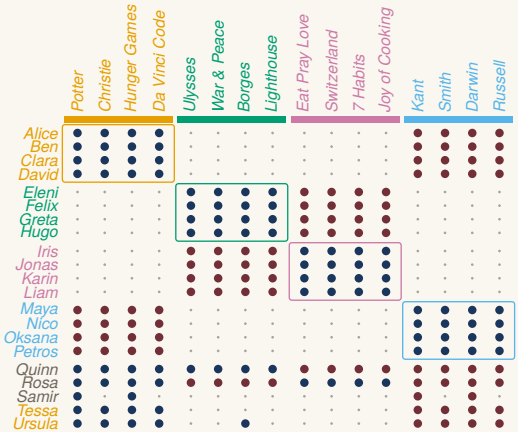


Three hundred and thirty-six cells. Can your brain remember them?

Now switching to:

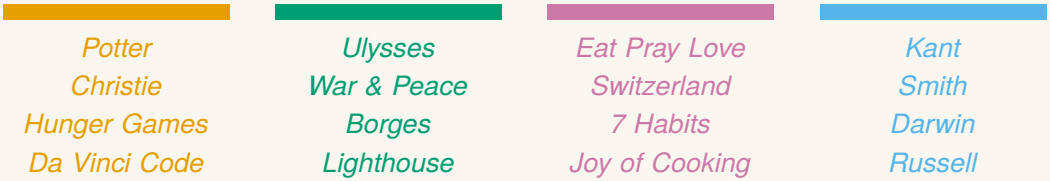
01_block-diagonal-puzzle.html

Sort the readers, sort the books



Sort the rows, sort the columns — four piles of books fall out.

Four piles of books. What is the pattern?



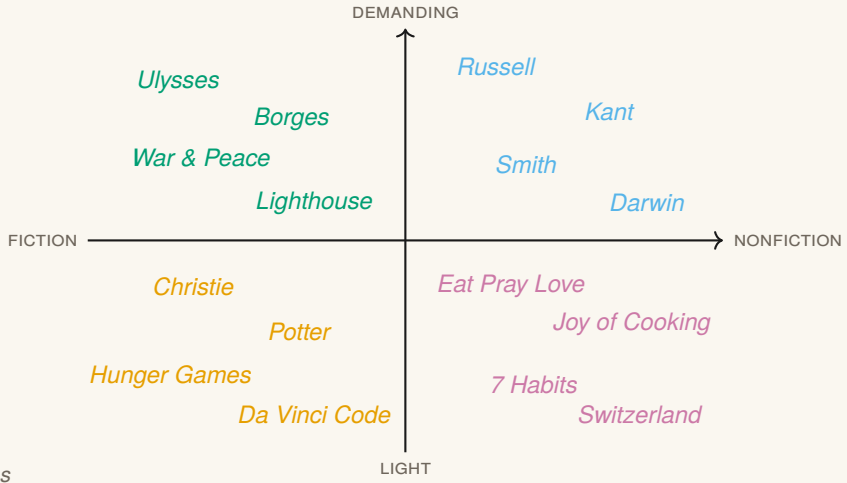
What two things does each pile share?

Sixteen books on one line



Sixteen books collapse to *two stacks*. Yet each stack carries *two colours* — one axis is not enough.

Add a second axis. Each book finds its corner.



The geometry was always there

Both axes are old library science: Dewey decimal (1876) gives Fiction vs. Nonfiction; Flesch (1948) and Lexile (1989) give Light vs. Demanding.

Two binary distinctions, four corners. This is how librarians and educators have sorted books for a century.

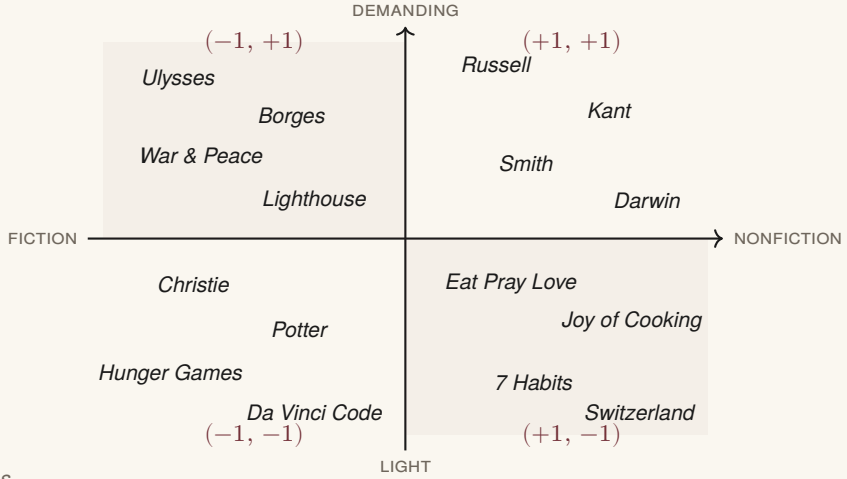
We did not impose these axes. We *recovered* them — from a matrix of likes and dislikes alone.

The books told us their geometry, and we listened.

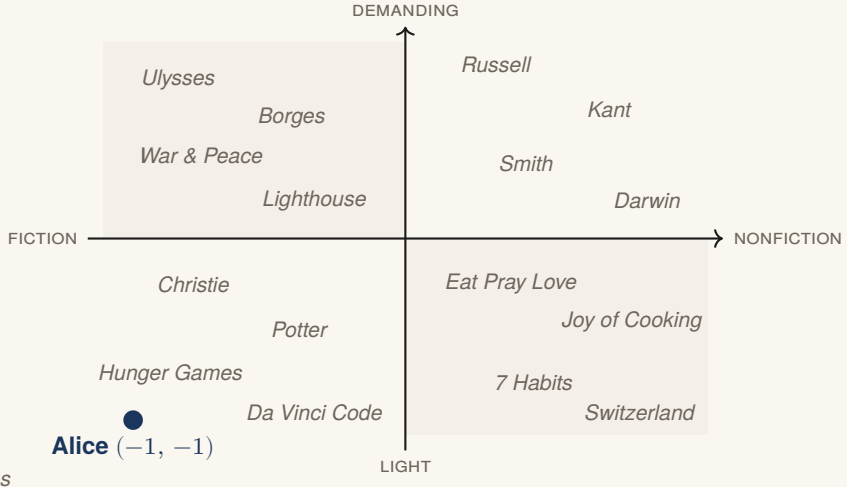
Now switching to:

02_latent-axes-lab.html

Each book gets two numbers

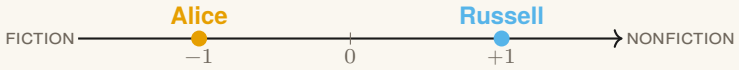


And readers, too



One axis at a time

Forget light vs demanding for a moment. Just fiction vs nonfiction.

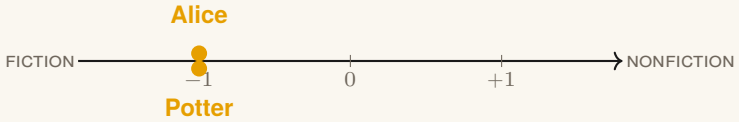


$$(-1) \times (+1) = -1$$

Different signs → −1 (a mismatch). Multiplication is a one-axis match score.

Same trick: Alice vs Potter

Same axis, different book.

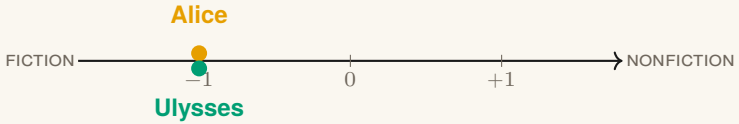


$$(-1) \times (-1) = +1$$

Same signs → +1 (a match). Both fiction; the axis agrees.

Same trick: Alice vs Ulysses

And again with Ulysses — a very different book.

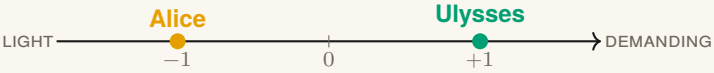


$$(-1) \times (-1) = +1$$

Both fiction — another match on this axis, even though Ulysses and Potter are nothing alike.

Now the light axis: Alice vs Ulysses

Same multiplication, second axis. Alice prefers *light* reads (-1); Ulysses is *demanding* ($+1$).



$$(-1) \times (+1) = -1$$

On this axis they disagree — Alice and Ulysses agree on fiction but not on demand.

Two axes, two votes: Alice vs Russell

Two votes per pair. *Add* them.

	Alice	**Russell**
	light fiction	demanding nonfiction
	$(-1,\ -1)$	$(+1,\ +1)$
FICTION-AXIS VOTE:		$(-1) \times (+1) = -1$
LIGHT-AXIS VOTE:		$(-1) \times (+1) = -1$
MATCH SCORE:		-2

× *Alice will not finish Russell.*

Two axes, two votes: Alice vs Potter

Same recipe. Different verdict.

	Alice	**Potter**
	light fiction	light fiction
	$(-1,\ -1)$	$(-1,\ -1)$
FICTION-AXIS VOTE:	$(-1) \times (-1) = +1$	
LIGHT-AXIS VOTE:	$(-1) \times (-1) = +1$	
MATCH SCORE:		$+2$

✓ *Alice will love Potter.*

Two axes, two votes: Alice vs Ulysses

Two votes that disagree.

	Alice	**Ulysses**
	light fiction	demanding fiction
	$(-1,\ -1)$	$(-1,\ +1)$
FICTION-AXIS VOTE:	$(-1) \times (-1) = +1$	
LIGHT-AXIS VOTE:	$(-1) \times (+1) = -1$	
MATCH SCORE:		0

? *A tie. Fiction matches; demand differs.*

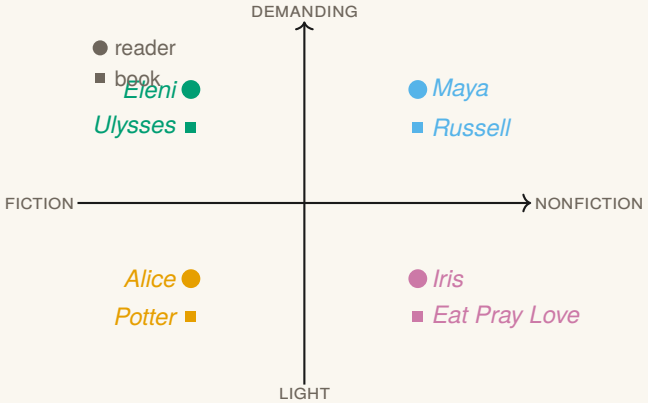
Same corner, opposite corner, sideways

READER \ BOOK	Potter	Ulysses	Eat Pray Love	Russell
Alice	$+2$	0	0	-2
Eleni	0	$+2$	-2	0
Iris	0	-2	$+2$	0
Maya	-2	0	0	$+2$

Same corner $\to +2$ (e.g. Alice $\times$ Potter). Opposite corner $\to -2$ (Alice $\times$ Russell). Side-by-side $\to 0$ (Alice $\times$ Ulysses).

Readers and books, same plane

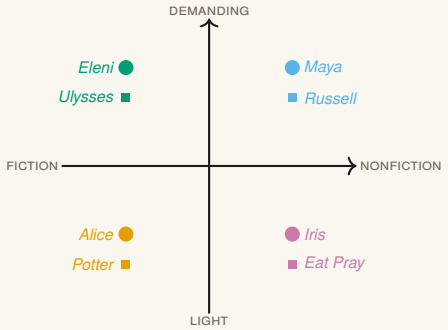
Each reader *and* each book has two coordinates — the same Cartesian plane carries them both.



The matrix arose from the embeddings

Each cell of the matrix is the inner product of two points in the plane: a *reader* times a *book*.

	Potter	Ulysses	Eat Pray	Russell
Alice	+2	0	0	−2
Eleni	0	+2	−2	0
Iris	0	−2	+2	0
Maya	−2	0	0	+2



Each matrix cell is one inner product of two points.

Preference is a dot product

The *match score* we have been computing has a name: the *dot product*.

$$r = u \cdot v$$

Multiply on each axis. Add the products.

For Alice meeting Russell: $u_{\text{Alice}} \cdot v_{\text{Russell}} = (-1)(+1) + (-1)(+1) = -2.$

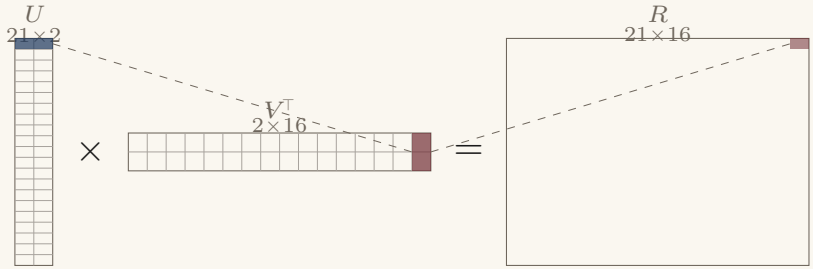
Two axes here, thousands in the wild

This lecture lives in 2 axes. Real recommenders live in tens to thousands. The formula does not change.

2 axes (today)	$u \cdot v = u_1 v_1 + u_2 v_2$
50 axes (a movie recommender)	$u \cdot v = u_1 v_1 + u_2 v_2 + \cdots + u_{50}\, v_{50}$
768 axes (word embeddings)	$u \cdot v = u_1 v_1 + \cdots + u_{768}\, v_{768}$
k axes (general)	$u \cdot v = u_1 v_1 + u_2 v_2 + \cdots + u_k\, v_k$

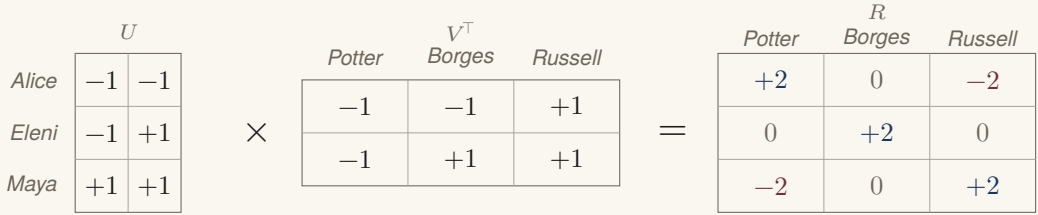
More axes, more terms. Same recipe: multiply on each axis, add.

Stack and multiply



21 readers in U, 16 books in V, their product *is* R. Each cell is one dot product — the highlighted one is Alice $\times$ Russell.

The matrix product, in numbers



Each entry of R is one dot product. *Positive* on the diagonal: each reader sits with her favourite book. *Negative* on the anti-diagonal: each reader far from the wrong book. *Zero* where the two are perpendicular.

But we only had R

We never observed U or V directly. The library's ledger has only R — the rating matrix, three hundred and thirty-six cells of symbols.

Yet the matrix is the product of two skinny ones. So the question is:

Given R alone, can we recover U and V?

This is the *matrix factorization* problem. The next demo gives one way to solve it.

Now switching to:

spectral-microscope.html

Eckart & Young (1936)[1]

> *Among all rank-k approximations of any matrix, the one given by the top-k singular vectors is optimal in squared error.*

No algorithm beats it. No clever trick beats it. No future architecture beats it.

The curve you just saw is not an empirical observation. It is the graph of a theorem ninety years old.

Now switching to:

customer-urns-carousel.html

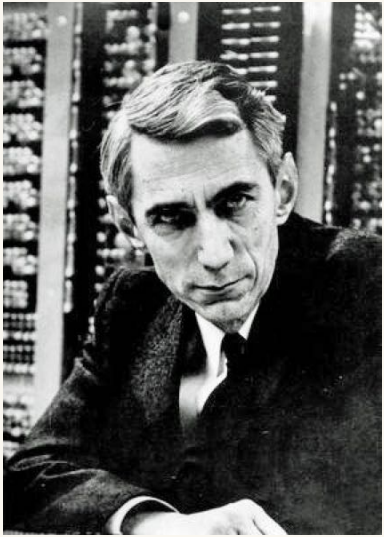
Now switching to:

07_embeddings-that-learn.html

End of Act I

We compressed a matrix. We named the act.

The names — *entropy*, *rank*, *factorization* — were old words from 1936 and 1948.



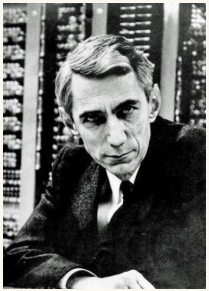
"The fundamental problem of communication is that of reproducing at one point either exactly or approximately a message selected at another point."

Claude E. Shannon, 1948

Interlude

Shannon, 1948

In 1948, Claude Shannon published *A Mathematical Theory of Communication* — a paper with two theorems at its core. The *source coding theorem* fixes a floor on how compactly any sequence of symbols can be described; the *channel coding theorem* fixes a ceiling on how reliably they can be transmitted through noise. For the next seven minutes we restate the first of these in the only vocabulary this lecture admits: urns.



How unpredictable is the next ball?

An urn with bias p produces a stream of white and yellow draws. When $p = \frac{1}{2}$ every draw is a coin flip — maximal surprise. When $p = 0.99$ almost every draw is white — the sequence is nearly redundant. *Entropy* is the name we give to this gradient between certainty and surprise. Let us feel it before we write it down.

Now switching to:

entropy-mountain.html

The entropy of an urn

$$H(p) = -p \log_2 p - (1-p) \log_2 (1-p)$$

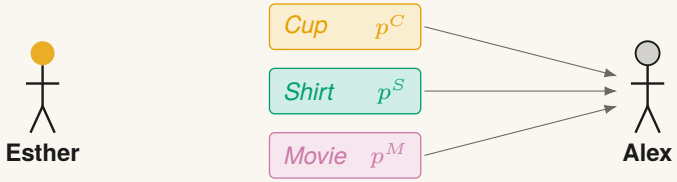
No compression scheme can describe an urn's draws in fewer than $H(p)$ bits per ball. Eckart–Young was the floor for compressing a matrix; entropy is the floor for compressing a sequence.

Recsys was compression. Bandits will be prediction.
Same urns, different question.

Act II — Prediction

Three campaigns

Esther runs an online shop. Three ad campaigns to test, one customer profile, a small budget.



1. How do we estimate p^C, p^S, and p^M?
2. How do we *efficiently* find which is the largest?

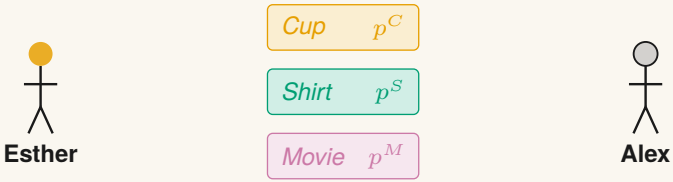
Applications

Bandits show up wherever you must *choose* from a menu of options whose payoffs you do not know.

- *Ad selection.* Which campaign converts most often?
- *Content recommendation.* Which post earns the next click?
- *Resource allocation.* Where does the next dollar buy the most return?
- *Robotics.* Which motion succeeds most reliably?

Same recipe, different names: explore, exploit, learn.

Challenges



1. Neither Esther *nor Alex* knows p^C, p^S, p^M in advance.
2. Showing an ad costs Esther money — she cannot test forever.

The solution: UCB algorithm

ALGORITHM 1. UPPER CONFIDENCE BOUND (UCB1)

Require: $\alpha > 2, \quad T \in \mathbb{N}$

1: $S_{0,i} = 0, \quad \hat{\mu}_{0,i} = 0, \quad i = 1, \dots, d$

2: **for** $t = 1$ **to** T **do**

3: $\quad A_t = \arg\max_{i=1,\dots,d} \left\{ \hat{\mu}_{t-1,i} + \sqrt{2\alpha \ln t \,/\, S_{t-1,i}} \right\}$

4: $\quad$ observe reward g_{t,A_t}

5: $\quad S_{t,i} \leftarrow S_{t-1,i} + \mathbf{1}[A_t = i]$

6: $\quad$ update $\hat{\mu}_{t,i} = \dfrac{1}{S_{t,i}} \sum_{j=1}^{t} g_{j,A_j} \mathbf{1}[A_j = i]$

7: **end for**

Read this — don't decode it. The rest of Act II builds this algorithm from scratch.

Agenda

1. *Estimating one probability.*

   Customers as urns.

2. *Multiple probabilities and multiple urns.*

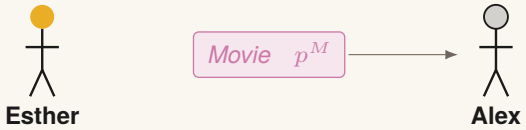
   ε-greedy and UCB.

3. *Many customers and items.*

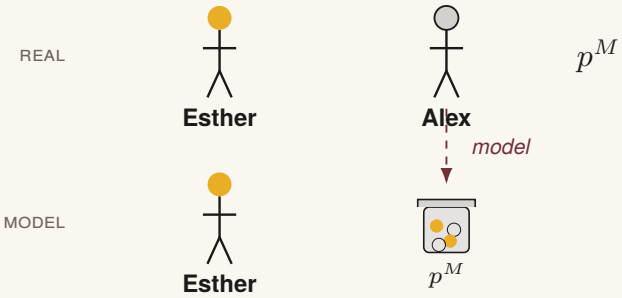
   Contextual urns.

Act I: Estimating one probability

Just one product. Just one probability.

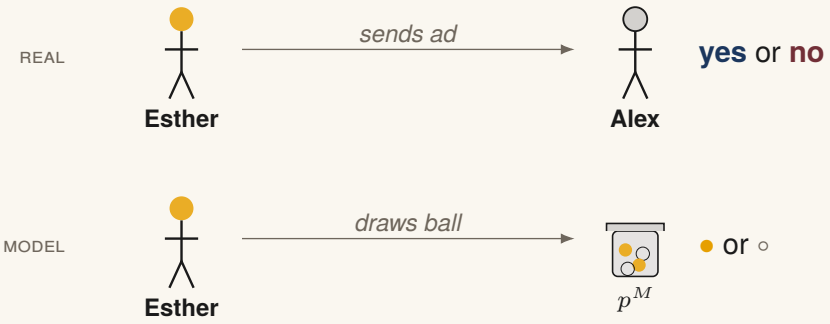


The customer is an urn



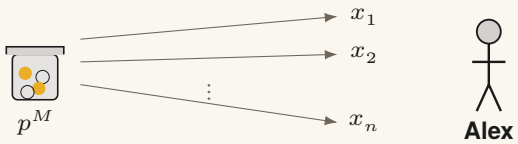
Alex's response distribution is the bias p^M. We model him as an urn whose balls are clicks and non-clicks.

The customer is an urn



$$p^M = \frac{w^M}{n^M}$$

The empirical estimator



($x_i = 1$ if conversion, 0 otherwise.)

$$\hat{p}_n^M \;=\; \bar{x}_n \;=\; \frac{x_1 + x_2 + \cdots + x_n}{n}$$

The empirical estimator converges

As n grows, $\hat{p}_n^M$ gets closer and closer to p^M with high probability.

$$\hat{p}_n^M \xrightarrow[n\to\infty]{\mathbb{P}} p^M$$

Formally, for any $\delta, \epsilon > 0$, there is $N \in \mathbb{N}$ such that

$$\mathbb{P}\big(|\hat{p}_n^M - p^M| > \delta\big) < \epsilon \quad \text{for } n \geq N.$$

This is an instance of the law of large numbers.

Confidence intervals

Estimating from few balls is risky.

Better: an *interval* instead of a point estimate.

A simple choice:

$$\left(\hat{p}_n - \tfrac{1}{\sqrt{n}},\ \hat{p}_n + \tfrac{1}{\sqrt{n}}\right)$$

sample	*interval*
$n = 5,\ \ \hat{p} = \frac{2}{5}$	$(\frac{2}{5} - \frac{1}{\sqrt{5}},\ \frac{2}{5} + \frac{1}{\sqrt{5}})$
$n = 50,\ \ \hat{p} = \frac{12}{50}$	$(\frac{12}{50} - \frac{1}{\sqrt{50}},\ \frac{12}{50} + \frac{1}{\sqrt{50}})$
$n = 500,\ \ \hat{p} = \frac{124}{500}$	$(\frac{124}{500} - \frac{1}{\sqrt{500}},\ \frac{124}{500} + \frac{1}{\sqrt{500}})$

Same urn, different sample sizes. As n grows, the interval shrinks like $1/\sqrt{n}$.

Now switching to:

three-urn-casino.html

Agenda

1. *Estimating one probability.*

   Customers as urns.

2. *Multiple probabilities and multiple urns.*

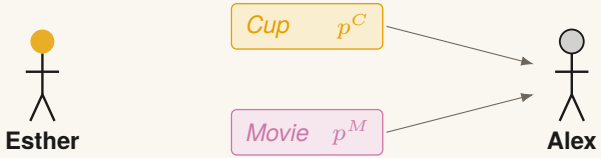
   ε-greedy and UCB.

3. *Many customers and items.*

   Contextual urns.

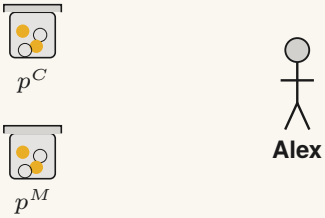
Act II: Two urns

Bring back the cup. Now two products, two probabilities.



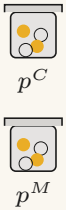
Naïve: estimate then exploit

Two urns, one customer, finite budget.



What is the simplest possible algorithm?

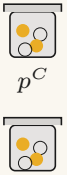
Naïve: estimate then exploit





1. Choose a number n.
2. Compute estimators $\hat{p}_n^C, \hat{p}_n^M$.
3. Exploit the urn with the largest estimate.

Drawbacks!





1. Choose a number n.
2. Compute $\hat{p}_n^C, \hat{p}_n^M$.
3. Exploit the largest.

- If p^C and p^M are close and n is too small, you exploit the *wrong urn*.
- If p^C and p^M are far apart and n is too large, you *waste rounds* estimating the loser.

ε-greedy: a first fix

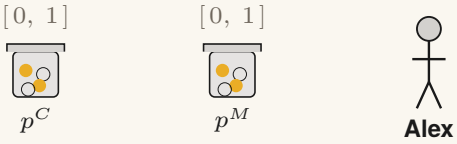
Don't commit. At each round, with probability:

- $1 - \varepsilon$: *exploit* the urn with the largest current estimate;
- ε: *explore* a random urn.

Most rounds, exploit. A small fraction, explore. The dial ε chooses between the two pressures.

Sutton & Barto, *Reinforcement Learning: An Introduction*, 1998/2018.

UCB sampling



1. For each urn u, start with $c_u = [0, 1]$.
2. For $t = 1, 2, \ldots$:
   - 2.1 find the urn $\hat{u}$ with the *highest upper bound*;
   - 2.2 draw a ball x from $\hat{u}$;
   - 2.3 if $x = 1$, get the reward;
   - 2.4 recompute the confidence interval of $\hat{u}$.

Race the algorithms

Four strategies, the same urns, the same hidden biases:

- Pure random — no learning at all.
- Pure greedy — exploit the leader from round one.
- ε-greedy — explore with probability ε.
- UCB — pick the urn with the highest upper bound.

Watch the cumulative-regret curves. The slope is the whole story: random and greedy grow regret *linearly*; ε-greedy and UCB grow it *sublinearly*.

Now switching to:

algorithm_stadium.html

Agenda

1. *Estimating one probability.*

   Customers as urns.

2. *Multiple probabilities and multiple urns.*

   ε-greedy and UCB.

3. *Many customers and items.*

   Contextual urns.

Act III: Many customers and many items

In a real recommender there are many customers and many items — too many urns to estimate one at a time.

1. Every item gets a representation vector z (e.g. from an autoencoder).
2. For every customer u, an unknown representation vector z_u.
3. The urn's bias is the inner product:

$$p(z, z_u) \;=\; \max\left(0,\; z^\top z_u\right).$$

4. Estimate z_u for each customer via regression.

Two acts, one geometry

Look back at Act I. Each reader had a vector u. Each book had a vector v. Preference was the inner product:

$$r \;=\; u \cdot v.$$

In Act III, each customer has a vector z_u. Each item has a vector z. The urn's bias is the same inner product:

$$p \;=\; \max(0,\; z^\top z_u).$$

Same matrix factorisation, two different settings.
Compression powers prediction.

How many users do you need?

It depends on the gap.

Two urns with conversion rates 0.50 and 0.40 are easy to tell apart. Two urns at 0.50 and 0.49 are nearly indistinguishable; separating them takes *far* more samples.

The required sample size scales roughly as $1/(p_1 - p_2)^2$. Halve the gap, and you need four times the data.

Lai & Robbins (1985)

$$\liminf_{T\to\infty} \frac{R_T}{\log T} \;\geq\; \sum_{i\neq i^*} \frac{\mu^* - \mu_i}{\mathrm{KL}(\mu_i \,\|\, \mu^*)}$$

No algorithm beats this rate. The hardest thing is telling apart distributions that look alike — and the cost is in KL divergence, the same Shannon used in 1948.

Two halves, one ceiling

Eckart–Young bounded compression.

Lai–Robbins bounds prediction.

The two halves of this lecture share one ceiling — and we have not yet named the quantity in which both ceilings are written. That is the Epilogue.



Epilogue — Duality

A perfect predictor is a perfect compressor

Two halves of one subject.

- A *forecaster* assigns $\hat{p}$ to the next symbol; her loss is $-\log \hat{p}$ per draw.
- An *encoder* writes short codewords for likely symbols; his cost is bits per draw.
- Shannon (1948) proved these are the same number, bottoming out at $H(p)$.

Hutter calls this *compression is comprehension*; Delétang et al. (2023) confirm it: a text-trained LM compresses ImageNet better than PNG.

Floor + inefficiency

$$\underbrace{-\,\mathbb{E}_p[\log \hat{p}]}_{\text{cross-entropy}} = \underbrace{H(p)}_{\text{Shannon's floor}} + \underbrace{\mathrm{KL}(p \,\|\, \hat{p})}_{\text{your inefficiency}}$$

Recsys: KL is the gap between your factorization and the truth. Bandits: KL is the cost of mistaking one urn for another. Compression: KL is the bits you waste because your predictor is imperfect. One quantity, three roles.

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